

NON-LINEAR EEG SIGNAL FEATURE EXTRACTION AND HEALTHY OR EPILEPTIC USER SEIZURE DISCRIMINATION BY WEIGHTED FREQUENCY BEHAVIOR ANALYSIS

Bruno Senzio-Savino^{*1}, Kyosuke Hirata¹, Mohammad Reza Alsharif¹, Kamaledin Setarehdan²

¹Department of Information Engineering, University of the Ryukyus, Okinawa, Japan

²School of Electrical and Computer Engineering, College of Engineering, University of Tehran, Iran

^{*}Corresponding author (Phone: + 81 (080) 64823582; b.senzio@gmail.com)

Abstract:

There are different types of seizures and they can happen for many different reasons, including physiological or psychological reactions or other body organ dysfunction. Epileptic seizures are caused by disturbances in the electrical activity of the brain. Although there are different preconditions in which a subject can have epileptic seizure activities, there is an alteration of the normal neural rhythms that can be detected by Electroencephalogram (EEG). The purpose of this research is to detect neural seizure activity on normal or preconditioned subjects at a non-invasive level with the use of EEG. In order to do that, a dataset consisting of EEG epochs from healthy and epileptic patients was analysed with the use of the Hilbert Huang Transform (HHT) and Intrinsic Mode Function (IMF) analysis with the use of weighted frequency components to analyse optimum classification component. The dataset consists of data samples that were obtained from epileptic patients during seizure and normal activity time frames both in epileptogenic and opposite area with the standard electrode 10-20 positioning system. Also, healthy user data of people with eyes open and closed was generated in the same areas. The dataset has been used by other researchers to analyse different characteristics. Instead of using a clustering algorithm to divide the obtained features as proposed by other authors, the analysis of weighted frequency behaviour on the IMF components could lead to the selection of the proper IMF components that were more suitable for a binary simple linear Support Vector Machine implementation that has a small processing time in comparison with other variants of this method. By applying WF behaviour analysis on all the IMFs, it is possible to reduce resourceful processing time that otherwise could be used by clustering algorithms. After decomposing by HHT and calculating the weighted frequency of all the five dataset available samples (100 data samples per dataset), from the weighted frequency behaviour analysis, two subsets were created: one subset consisting epileptic and epileptic seizure samples (300 total) and another subset containing healthy and epileptic seizure data (300 total). From an 80-20 linear Support Vector Machine implementation, the recognition rates of the first and second subset were of 93.3% and 86.7% respectively. The use of this method allows a low processing resource implementation over different platforms. Currently, a cloud based seizure recognition platform will be implemented for the current data and other seizure activity might also be introduced in the near future.

Keywords: Support Vector Machine, Hilbert Huang Transform, Weighted Frequency, EEG Signal Processing, Intrinsic Mode Function

1. HEALTHY / EPILEPTIC EEG SEIZURE SIGNAL CLASSIFICATION BACKGROUND

In the past, there has been different ways of classifying EEG signals for different purposes. Complex signals like the EEG are of massive volume and require the personal judgment of an expert for analysis. Some authors referred the use of Artificial Neural Networks for seizure detection [1] and use different signal analysis methods like signal entropy [2]. Other techniques include the use of other Artificial Intelligence and Machine Learning methods. One of the most used classification methods for EEG signal analysis employs Linear Discriminant Analysis based techniques like Support Vector Machines. Although there are different ways (kernels) to classify data by this method, the one that uses the lowest resources is based on a simple linear kernel by finding an optimal hyper plane. This technique has been proven to have good classification results when used on Brain Computer Interface [3]. However, it is important to notice that simple linear kernel classification can be disrupted in a high sense when the presence of noise or cluster outliers is high. Therefore, the need of finding a good set of feature vectors for classification becomes necessary. In this research, a method for nonlinear feature vector extraction is analyzed and a new way for analyzing proper samples is introduced. The use of this method was used to simply classify and analyze epileptic, healthy and seizure activity from a set of samples.

1.1. HEALTHY / EPILEPTIC EEG SIGNAL DATASET DESCRIPTION

For this research, in order to perform the required classification, it was necessary to obtain reliable data that could allow understanding the signal nature of both healthy and ictal (seizure) activities from different subjects. One of the freely available sets [5] consisted of five datasets (denoted A-E) containing 100 single channel EEG segments of 23.6 second duration. The device sampling frequency was of 173.61 Hz and Band-pass filter of 0.53-40Hz was applied as a preprocessing stage. The datasets consisted of data from both healthy and epileptic subjects from electrodes positioned on non-epileptogenic (for the preconditioned subjects) and normal scalp regions [4]. Healthy volunteers with eyes open (dataset Z) and closed (O) as well as non-seizure opposite hippocampus (N) as well as epileptogenic (F) and ictal activity (S) information was used for this research analysis.

2. HILBERT-HUANG TRANSFORM (HHT) AND WEIGHTED FREQUENCY (WF)

In order to perform nonlinear feature vector extraction from the original data, the Hilbert Huang Transform is commonly used to decompose a signal into different components by means of the Empirical Mode Decomposition (EMD), which is an essential part for the Hilbert Spectral Analysis.

EMD consists of a sifting process that obtains the lower and higher boundaries of a signal in order to decompose them with the use of its mean [5]. The process stops when the signal variance is low or by a predefined parameter. For this research purpose, the sifting process stops when the predefined parameter goes below 0.1. Once the sifting process stops, all the generated signals are called Intrinsic Mode Functions (IMF). Figure 1 presents the EMD of some dataset samples.

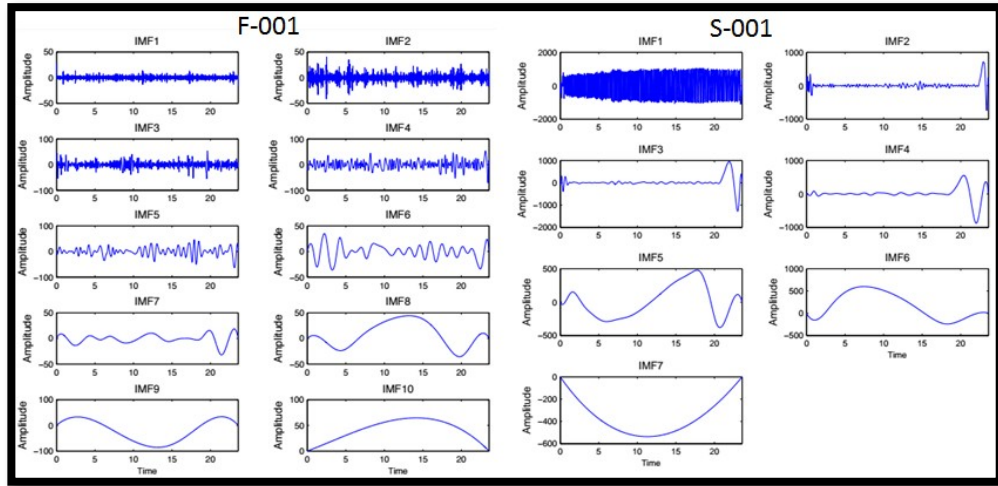


Figure 1. EMD of some F and S subset samples

In order to obtain measure the rate of frequency changes for a signal, the use of the Hilbert frequency analysis results useful. This analysis consists of obtaining an analytical signal from the original one by means of the Hilbert transform $y(t)$ [6], which consists of:

$$y(t) = \frac{1}{\pi} \left(\int_{t-\frac{1}{\epsilon}}^{t-\epsilon} \frac{x(\tau)}{t-\tau} d\tau + \int_{t+\frac{1}{\epsilon}}^{t+\epsilon} \frac{x(\tau)}{t-\tau} d\tau \right) \quad (1)$$

The analytic signal is obtained by the following arrangement

$$z(t) = x(t) + iy(t) = a(t) e^{i\theta(t)} \quad (2)$$

Where instantaneous amplitude is defined by:

$$a(t) = \sqrt{(x(t))^2 + (y(t))^2} \quad (3)$$

And instantaneous phase is obtained with:

$$\theta(t) = \arctan \left(\frac{y(t)}{x(t)} \right) \quad (4)$$

The instantaneous frequency of the analytic signal is obtained by a derivation of the phase function as follows:

$$f(t) = \frac{1}{2\pi} \frac{\delta \theta(t)}{\delta t} = \frac{\omega(t)}{2\pi} \quad (5)$$

Once the instantaneous frequency is calculated, this research used weighted frequency proposed by [6] to separate signal components as explained later. The weighted frequency of gives idea of the mean frequency using instantaneous information [6]:

$$\bar{f} = \frac{\sum_{i=1}^k a(i) (f(i))^2}{\sum_{i=1}^k a(i) f(i)} \quad (6)$$

For this research, due to the numerical methods used for the derivative function of the phase function, some steps provided negative values. However, to simplify the processing time, these values were set to zero in order to avoid miscalculations. Also, it is noted that this arrangement does not affect the overall weighted frequency calculation since it is applied equally for all the dataset samples.

2.1. WEIGHTED FREQUENCY BEHAVIOR ANALYSIS

In previous seizure classification research based on the same approach [6], clustering by k-means took an important role for feature classification of data. However, for this case, an analysis of the different weighted frequency values was realized in order to observe and determine the set of components (IMF values) that could represent the best separation for a binary classification task by means of simple linear Support Vector Machine (SVM) classification. From the obtained data, it was possible to analyse the weighted frequency values of all the generated IMF components from all datasets. However, those who presented a better separation possibility were the first and sixth IMF components of the data. Figure 2 presents a couple weighted frequency plots of the performed calculations for all subset samples. Since the purpose of this research is to separate classification between healthy/seizure and epileptic/seizure signal activity, this analysis also provided a way to inspect the data in such a way to allow the proper subset division.

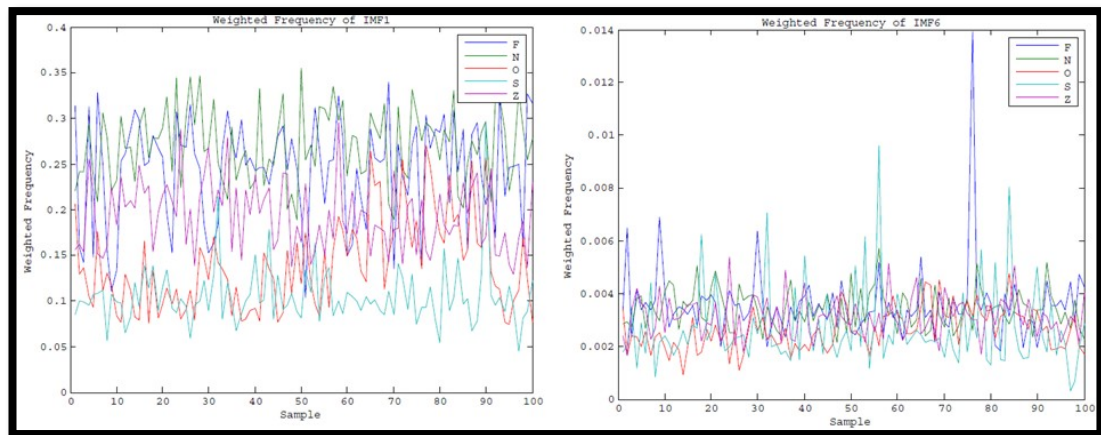


Figure 2. Weighted frequency behaviour analysis of different subsets

3. RESULTS

From the weighted frequency behaviour analysis, the components that provided a better classification for the simple linear SVM classification were the first and sixth components (WF_IMF1 and WF_IMF6) of the data. The resulting classifiers are shown on Figure 3.

In order to classify the data and perform recognition rate calculations, from the current samples, an 80-20 (80% training and 20% test) basis classifier was built. The results are presented on Table 1.

Table 1. SVM Classification Recognition Rates

Classifier	Recognition rate
Epileptic/Seizure	93.3%

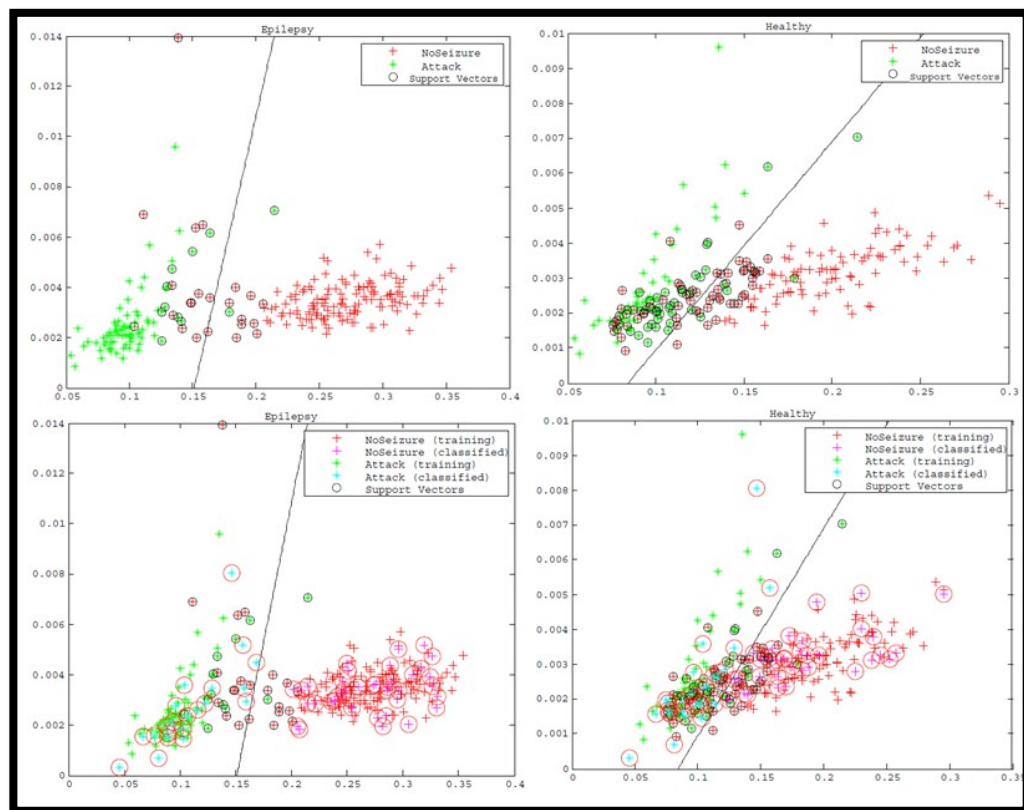


Figure 3. The resulting SVM for Epileptic/Seizure and Healthy/Seizure datasets (Top: Train, Bottom: Test)

4. CONCLUSIONS AND FURTHER WORK

The classification rates for the sample data was high and a classification for both conditions could be achieved successfully. The methodology proposed in this research allows to save a clustering stage, which can be necessary when task processing times have to be taken into account. The implementation of a cloud service in charge of processing the data via a remote classifier is currently being developed and will be finished in the near future.

References:

- [1] N. Pradhan, P.K. Sadasivan and G.R. Arunodaya.: Detection of Seizure Activity in EEG by an Artificial Neural Network: A Preliminary Study. Computers and Biomedical Research, p.303-313(1996)
- [2] V. Srinivasan, C. Eswaran and N. Sriraam. Approximate Entropy-based Epileptic EEG Detection Using Artificial Networks. IEEE Transactions on Information Technology in Biomedicine, p.288-295 (2007)
- [3] F. Lotte, M. Congedo, A. Lecuyer, F. Lamarche and B. Arnaldi. A review of classification algorithms for EEG-based brain-computer interfaces, Journal of Neural Engineering, pp.24 (2004).
- [4] Andrzejak RG, L. K. Rieke C, Mormann F., David P and Elger CE. Indications of nonlinear deterministic and finite dimensional structures in time series of brain electrical activity: Dependence on recording region and brain state, Phys. Rev. E, 4, 061907 (2001).

[5] J. Otsuka. Hilbert-Huang Transform による非定常・非線形信号分析, 技術資料, 寒地土木研究所月報, pp.39-44 (2011).

[6] R. J. Oweis and E.W. Abdulhay. Sizable classification in EEG signals utilizing Hilbert-Huang Transform, BioMedical Engineering Online (2011).